

PCU-Select: Amortizing Data Selection Across a Registry of PEFT Configurations

Anonymous submission

Abstract

A machine-learning platform often maintains several parameter-efficient fine-tuning (PEFT) configurations for different tasks and deployment budgets. Gradient-based data selectors preserve target specificity, but they rebuild a gradient datastore whenever the PEFT configuration changes. Reusing one subset avoids this computation but loses quality because different trainable subspaces value different examples. We present PCU-Select, which amortizes data selection across a PEFT registry. PCU-Select represents every supported PEFT family at 24 shared intervention sites: Transformer outputs through which PEFT updates affect the model. It caches reusable sample and task signals at these sites, then combines them with a code that describes the target PEFT. After a one-time offline stage, a single scorer ranks candidates for any supported PEFT-task pair without rebuilding gradients. On Llama-2-7B across four tasks and five seen PEFTs, PCU-Select achieves 35.18 average quality, nearly tying per-PEFT LESS at 35.14. Onboarding a new four-task PEFT costs 1.44 GPU-hours instead of 63.2, a 44-fold reduction. PCU-Select thus preserves the quality of per-PEFT selection while reducing its marginal cost at registry scale.

1 Introduction

Efficient adaptation of large language models requires reducing both the number of trainable parameters and the amount of training data. A machine-learning platform must also serve heterogeneous tasks and deployment constraints, and therefore rarely relies on a single adaptation setting. Instead, it keeps a *registry* of PEFT configurations for different tasks, latency targets, and memory limits. The registry may include LoRA at several ranks and placements, adapters, (IA)³, prefix tuning, and BitFit. Data-efficient fine-tuning complements these methods by selecting a compact subset from a larger instruction pool. In current practice, however, PEFT controls which parameters move while data selection independently chooses which examples drive the update.

Current data selectors rarely model this interaction. Quality filters, diversity heuristics, and proxy-model scores assign each example one value for a target task, then reuse the selected subset across adaptation methods. This PEFT-agnostic

assumption ignores how each method changes the trainable subspace. LoRA injects low-rank additive updates into selected projections (Hu et al. 2022); adapters insert residual bottlenecks (Houlsby et al. 2019); and (IA)³ learns multiplicative activation scalings (Liu et al. 2022). An example aligned with a task inside one subspace need not remain aligned inside another.

When one pool must serve a whole registry, this omission becomes a cost problem. LESS (Xia et al. 2024), a strong gradient-based selector, builds a target-specific gradient datastore. It then ranks candidates by their influence on a *task sketch*, a small validation set that represents the target task. Because the datastore depends on the target adapter’s trainable parameters, each configuration must rebuild it. Selection cost therefore grows linearly with the number of PEFTs. The obvious shortcut selects once and reuses the subset everywhere. It removes the recomputation cost but also loses the target specificity that makes gradient selection valuable. Indeed, reusing a single LESS subset performs worse than a PEFT-agnostic representation baseline (Table 2). Existing methods thus face a reuse-versus-recompute dilemma: they are either expensive for every target or effectively PEFT-agnostic.

Our motivation study shows that PEFT-specific utility changes with the trainable subspace. Figure 1 scores one candidate pool under LoRA, adapter, and (IA)³ variants. Independent replicates under the same configuration agree more strongly than scoring runs across configurations. Same-family changes preserve partial structure, whereas cross-family changes produce the largest drop in rank agreement and Top-5% overlap. The transfer matrix shows the downstream consequence. Selecting data for the training PEFT beats mismatched *cross-family* selection by 2.0 points on average across target PEFTs (Appendix Table 12). The Analysis section reports a narrower 1.45-point gap within a LoRA configuration scan.

This paper treats data value as a PEFT-conditioned quantity. Given an example x , task t , and PEFT configuration p , we predict whether training on x will help task t . This prediction is specific to the trainable subspace defined by p . PEFT-agnostic selection asks only whether x is generally useful for the task. Per-PEFT influence selection captures the configuration but recomputes expensive supervision for every target. Our formulation retains the configuration con-

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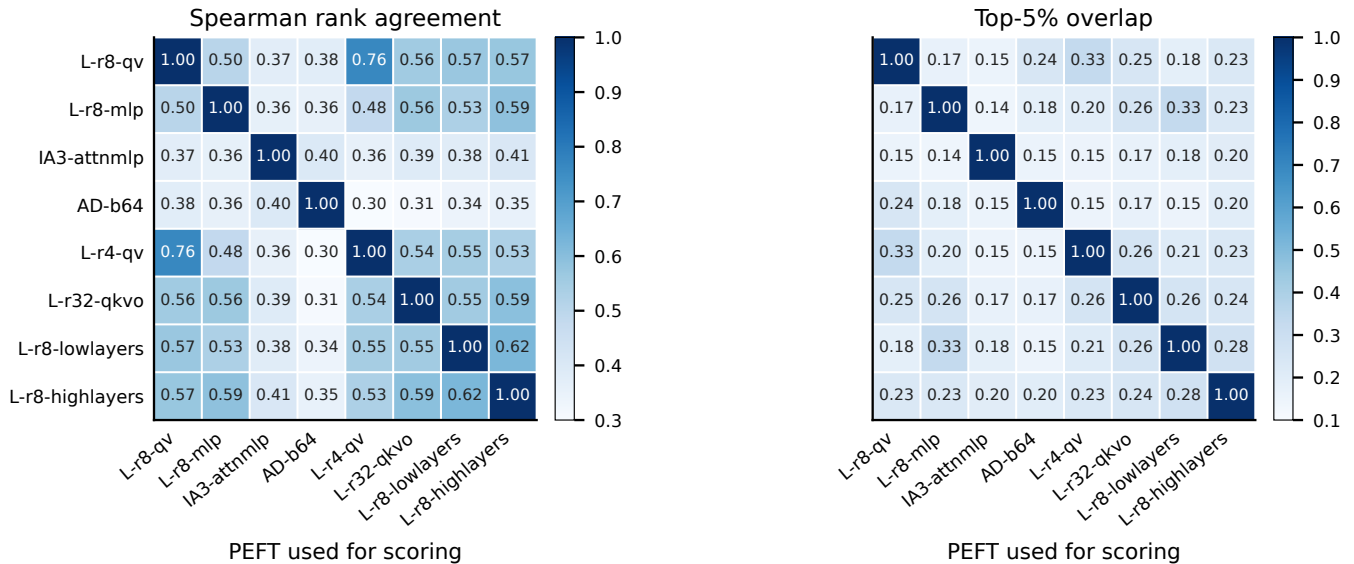


Figure 1: Short-update data-value estimates disagree across PEFT configurations. We score a common pool of 20K candidate examples under each target PEFT using two task sketches and independent anchor/seed replicates. Each cell reports the Spearman rank correlation between its row and column runs. A run compared with itself yields the 1.00 diagonal. Independent runs for the same PEFT agree only moderately ($\rho = 0.79$, Top-5% overlap = 0.32), which establishes the noise floor of the short-update labels. Agreement falls further as PEFT structures diverge: cross-family pairs average $\rho = 0.36$ and Top-5% overlap = 0.17.

dition while sharing supervision across targets.

PCU-Select (PEFT-Conditional Utility) separates reusable signals from target-specific decisions. It maps every supported PEFT family onto 24 *intervention sites*: selected Transformer outputs through which PEFT updates affect the model. Samples and task sketches share this coordinate system, so PCU-Select can cache their signals once. A structured PEFT code then specifies where and how the target configuration acts. PCU-Select trains one conditional scorer from two label sources. Cheap gradient-proxy labels cover the candidate pool, while short-update labels correct consequential ranking errors. At deployment, the scorer combines the cached signals with the target code, ranks the pool, and spreads the budget across semantic clusters to avoid near-duplicates.

This design amortizes target-specific selection across a PEFT registry. Our contributions are three-fold:

- **PEFT-conditioned utility.** We formulate data value as $u(x, p, t)$ and show that both data-value rankings and the downstream quality of selected subsets depend on the PEFT configuration (Figure 1; Appendix Table 12).
- **Intervention-site amortization.** PCU-Select represents incompatible PEFT parameterizations through the same 24 module-output sites. It shares sample and task signals across configurations, while a structured PEFT code preserves target specificity without rebuilding a gradient datastore. Removing the PEFT code or task sketch reduces downstream performance by 1.37 or 0.82 points, respectively (Appendix Table 11).
- **Registry-scale quality and cost.** Across four tasks and five seen PEFTs on Llama-2-7B (Touvron et al. 2023),

PCU-Select gains 0.74 points over RDS+ and differs from per-PEFT LESS by +0.04. It reduces the selection-stage cost per added four-task PEFT by a factor of 44 (1.44 versus 63.2 GPU-hours; Tables 1 and 2).

2 Related Work

Instruction-data selection. Instruction tuning benefits from small, carefully curated subsets. LIMA demonstrates that 1,000 curated examples produce strong instruction-following behavior in a large model (Zhou et al. 2023). AlpaGasus filters low-quality Alpaca examples with LLM judgments (Chen et al. 2024); InsTag measures semantic diversity and complexity through instruction tags (Lu et al. 2023); and DEITA combines quality, complexity, and diversity criteria for automatic data selection (Liu et al. 2024). Other selectors emphasize explicit diversity through quality-diversity trade-offs, log-determinant objectives, or partition-aware sampling (Bukharin et al. 2023; Wang et al. 2024; Rao et al. 2024). These methods improve data efficiency, but they mostly treat value as a property of the example and target task rather than of the PEFT subspace used for adaptation.

Proxy-model and influence selection. Model-assisted selectors score examples with external judges, weak models, or training signals. Cherry LLM uses instruction-following difficulty (Li et al. 2023); Superfiltering studies when weak models can rank data for stronger models (Li et al. 2024); and SmallToLarge transfers trajectory information from small models to larger targets (Yang et al. 2024).

Influence-style methods instead estimate how training examples affect validation behavior. Classical influence functions use second-order approximations (Koh and Liang

2017). TracIn accumulates gradient similarities along checkpoints (Pruthi et al. 2020), LESS builds a low-dimensional gradient datastore for instruction selection (Xia et al. 2024), and DataInf specializes influence approximation for LoRA-tuned generative models (Kwon et al. 2024). These gradient methods derive utility from one target parameterization. Preserving PEFT specificity therefore forces recomputation whenever the configuration changes. Cheap quality, semantic, and weak-model signals avoid that cost but ignore the PEFT subspace. PCU-Select resolves this reuse-versus-recompute dilemma by conditioning one scorer on a structured PEFT code. We compare it against both a per-PEFT LESS reimplementaion and a cheaper Influence baseline with one shared datastore.

Parameter-efficient fine-tuning. PEFT methods reduce adaptation cost by updating a small module or parameter subset. Adapters insert bottleneck modules into Transformer layers (Houlsby et al. 2019; Pfeiffer et al. 2021). Prefix and prompt tuning optimize continuous prompts or prefix states (Li and Liang 2021; Lester, Al-Rfou, and Constant 2021). LoRA learns low-rank additive updates (Hu et al. 2022), QLoRA combines quantization with LoRA for memory efficiency (Detmiers et al. 2023), BitFit tunes only bias terms (Ben Zaken, Goldberg, and Ravfogel 2022), and (IA)³ rescales attention and feed-forward activations (Liu et al. 2022). PEFT work usually compares accuracy, parameter count, memory, or inference overhead under fixed data. This paper asks a complementary question: when the candidate pool is fixed, does example utility remain stable across PEFT configurations?

Joint data- and parameter-efficient adaptation. Data selection and PEFT address two sides of efficient adaptation, but most work optimizes one side while holding the other fixed. DataInf and LESS use gradients from LoRA-style training (Kwon et al. 2024; Xia et al. 2024). Recent geometry-aware targeted selection also shows that PEFT optimization has nontrivial subspace structure (Min et al. 2026). Their utility estimates remain tied to the optimization geometry of a particular target configuration. PCU-Select instead conditions on the *structured PEFT configuration itself*: its sites, operator, capacity, and recipe. This representation shares supervision across configurations without erasing their differences. The resulting requirement is precise: a selector must model PEFT structure and task-relative utility while reusing its expensive computations. We formalize that requirement next.

3 Problem Formulation

We formalize *cross-PEFT data-efficient fine-tuning*: selecting a compact training subset whose value depends on both the target PEFT configuration and the task. Fix a backbone family and a frozen base model M_0 . A candidate pool $\mathcal{D} = \{x_i\}_{i=1}^N$ contains instruction-response examples. Each task t provides a small task sketch V_t from its train or development data, never from its test set. Each PEFT configuration $p \in \mathcal{P}$ specifies a family, trainable locations, capacity, and

training recipe. Given budget B , the ideal subset solves

$$\mathcal{S}^* = \arg \max_{\mathcal{S} \subseteq \mathcal{D}, |\mathcal{S}| \leq B} \text{Perf}(\text{Tune}(M_0, p, \mathcal{S}), V_t^{\text{test}}). \quad (1)$$

The test set V_t^{test} appears only in this evaluation oracle. Every training, scoring, and selection procedure uses only the task sketch V_t . Directly optimizing Eq. (1) would require training and evaluating many subsets for every target (p, t) pair, which is infeasible.

PCU-Select replaces the subset oracle with a conditional per-example utility. It learns the surrogate

$$s_\phi(x, p, t) \approx u(x, p, t), \quad (2)$$

Here $u(x, p, t)$ denotes the utility of training on x for task t under the trainable subspace induced by p . Concretely, $u(x, p, t) \in [0, 1]$ summarizes loss reductions after a few update steps. PCU-Select rank-normalizes these reductions within condition-matched buckets (§4). Conditioning on p is the core departure from PEFT-agnostic selection: a change in the trainable subspace may change the selected subset.

Utility is relative to the task as well as the configuration. A worked-solution arithmetic example scores high under the GSM8K sketch but low under TyDiQA, and its utility shifts again across PEFTs that do not train the aligned sites.

Amortization objective. A registry makes total cost, together with quality, the operative criterion. Let P denote the number of target PEFT configurations and Q the number of task sketches served from a fixed pool. LESS builds one datastore per PEFT and ranks it against each task. PCU-Select instead pays one shared offline cost and one online scoring-and-selection pass per PEFT-task pair:

$$\begin{aligned} C_{\text{LESS}}(P, Q) &= P(C_{\text{LESS}}^{\text{data}} + QC_{\text{LESS}}^{\text{rank}}), \\ C_{\text{PCU}}(P, Q) &= C_{\text{offline}} + PQ C_{\text{PCU}}^{\text{pair}}, \end{aligned} \quad (3)$$

The three terms separate reusable work from target-specific work. $C_{\text{LESS}}^{\text{data}}$ builds one gradient datastore for a PEFT, and $C_{\text{LESS}}^{\text{rank}}$ ranks it for one task. $C_{\text{PCU}}^{\text{pair}}$ runs one online scoring-and-selection pass. This online pass costs much less than rebuilding a LESS datastore. We call the target *quality-preserving amortization*. It reuses expensive work across configurations while keeping the quality gap from a per-PEFT selector within a small tolerance ϵ :

$$\Delta_{\text{quality}} = \text{Perf}_{\text{PCU}} - \text{Perf}_{\text{LESS}}, \quad |\Delta_{\text{quality}}| \leq \epsilon, \quad (4)$$

It also reduces total cost once P exceeds $P^* = C_{\text{offline}} / [C_{\text{LESS}}^{\text{data}} + Q(C_{\text{LESS}}^{\text{rank}} - C_{\text{PCU}}^{\text{pair}})]$. The objective is LESS-level quality with a shared offline investment that replaces repeated per-configuration computation.

Design requirements. The formulation yields four requirements. **PEFT conditioning** makes p an explicit scorer input. **Task relativity** ties utility to the task sketch V_t , preventing the scorer from learning only generic example quality. **Structural representation** encodes intervention sites, operator type, capacity, and training recipe instead of using only a family name. **Amortizable execution** moves expensive label construction and scorer training offline, so a new

(p, t) pair uses cached features and forward scoring. PCU-Select meets these requirements with shared module-output coordinates, task-relative labels, and a PEFT-conditioned scorer. The next section turns them into a reusable selection pipeline.

4 Method: PCU-Select

PCU-Select learns the surrogate in Eq. (2) and uses it to select a budgeted subset for a target PEFT and task. Figure 2 shows two stages. Offline, PCU-Select extracts reusable sample and task signals, constructs gradient-proxy and short-update labels, and trains the scorer. Online, it encodes a target (p^*, t^*) and measures its structural support from the training registry. It then scores the pool and allocates the budget across semantic clusters. Algorithms 1 and 2 in the appendix specify both stages.

Shared Intervention Sites Enable Cross-PEFT Reuse

PEFT families expose parameter tensors with different shapes and roles, which prevents a shared raw-gradient cache. PCU-Select instead compares them at *intervention sites*: selected module outputs through which PEFT updates affect the backbone. In plain terms, a site records how a sample or task pushes an intermediate activation. This signal remains comparable across PEFT parameter shapes.

PCU-Select hooks attention, MLP, and post-residual outputs at eight evenly spaced Transformer layers, giving $|\Omega| = 24$. By the chain rule, PEFT-parameter gradients factor through the upstream gradients at the module outputs they modify. A site gradient therefore provides a shared proxy coordinate without reconstructing any full PEFT-parameter gradient. These coordinates avoid shape mismatches between fused and split projections. The post-residual coordinate also places adapters alongside LoRA and (IA)³.

The site mask records *where* a configuration acts. Separate operator and capacity features record *how* strongly and by what mechanism it acts. This factorization lets all configurations reuse one sample signature without treating additive, multiplicative, and bottleneck updates as interchangeable.

Each PEFT configuration maps to a site mask and an operator type. Attention LoRA and (IA)³-attention touch attention-output sites; MLP LoRA and (IA)³-FFN touch MLP-output sites; adapters touch block-residual sites; and prefix tuning touches attention-output sites. The code represents BitFit as a bias shift. PCU-Select assigns each site a normalized PEFT weight

$$\begin{aligned} w_p^\omega &= \text{mask}(p, \omega) \kappa(p, \omega), \\ \kappa(p, \omega) &= \tanh(\eta \log(1 + \text{cap}(p, \omega)/d_{\text{model}})), \\ \tilde{w}_p^\omega &= w_p^\omega / \max(\varepsilon_w, \sum_{\omega'} w_p^{\omega'}). \end{aligned} \quad (5)$$

Untouched sites receive zero weight, while capacity raises the weight of touched sites. Appendix Table 10 maps LoRA rank, adapter width, (IA)³ vector count, prefix length, and BitFit bias count to a trainable-parameter scale. The tanh limits the influence of high-capacity configurations. PCU-Select

encodes operator type separately in z_p because a family-wide scalar would cancel under the normalization in Eq. (5).

Shared Representations Preserve Target Specificity

The three representations answer different questions: what the example contains, what the task needs, and where the PEFT can act. PCU-Select expresses all three in the shared site coordinate system before constructing utility labels.

The sample representation $z_x = [e_x; d_x; a_x] \in \mathbb{R}^{848}$ combines semantic content, difficulty, format, and activation behavior. Its 768-dimensional semantic vector is the frozen `sentence-transformers/all-mpnet-base-v2` embedding of the instruction and response (Song et al. 2020; Reimers and Gurevych 2019). We do not train this encoder on the candidate pool or target benchmarks. A 16-dimensional vector records length, response-LM loss, entropy, and format features. A 64-dimensional activation signature summarizes response-token hidden states at the eight sampled layers.

The task representation $z_t \in \mathbb{R}^{848}$ describes what the target task needs. PCU-Select mean-pools z_x over the 32 examples in the task sketch. It also averages their per-site gradients into g_t^ω for the gradient-proxy label. These 24×256 gradients remain separate from z_t .

The PEFT representation $z_p = [m_p; c_p; r_p] \in \mathbb{R}^{128}$ describes where and how the target configuration acts. A 24×4 site/operator mask (m_p , 96 dimensions) marks active sites, additive updates, multiplicative scaling, and prefix-style control. Two 16-dimensional vectors encode capacity (c_p) and the training recipe (r_p). At the 24-site resolution, q/v and q/k/v/o LoRA touch the same attention-output sites. The capacity and recipe vectors preserve their projection-level distinction.

Prefix, P-Tuning, and BitFit use the same deterministic site, operator, capacity, and recipe rules as the seen families. The model uses only these structural features; it has no learned family lookup or operator centroid. Here, *zero-shot* means scoring a new configuration without target-specific short-update labels. When a PEFT family provides such labels, the residual head described below can calibrate the score.

PCU-Select trains from two complementary label sources. Gradient-proxy labels cheaply cover the full candidate pool. Short-update labels measure the effect of actual PEFT updates on fewer examples. Together they provide broad ranking structure and correct it near consequential selection decisions.

The *gradient-proxy label* measures alignment between each sample and the task sketch at sites used by the target PEFT:

$$u^{\text{lo}}(x, p, t) = \sum_{\omega \in \Omega} \tilde{w}_p^\omega \cos(g_x^\omega, g_t^\omega). \quad (6)$$

Here g_x^ω derives from the response-token loss gradient at site ω . A unit-normalized Johnson–Lindenstrauss random projection (Johnson and Lindenstrauss 1984) compresses it to a fixed length while approximately preserving its geometry. The sketch mean gives g_t^ω . All task sketches use response-token cross-entropy for these signatures. PCU-Select reserves task-native metrics such as option likelihood, Pass@1,

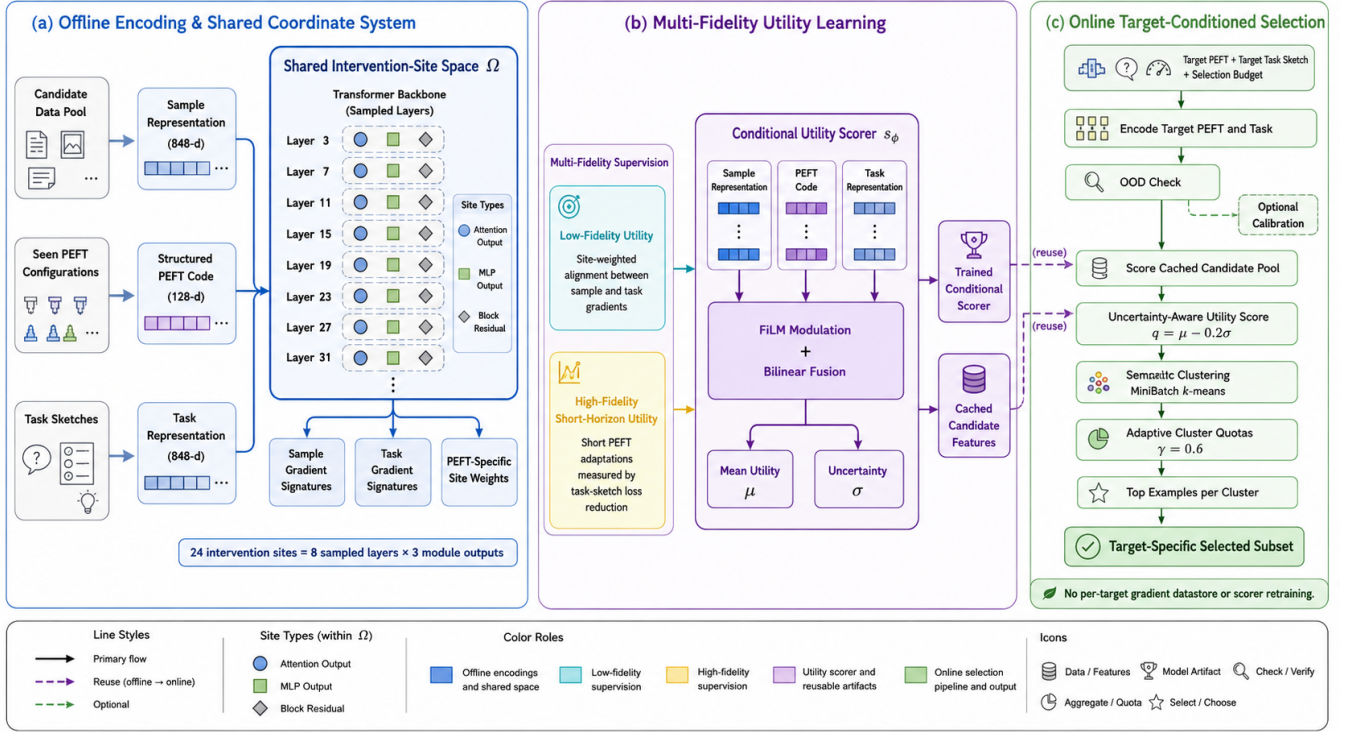


Figure 2: PCU-Select separates reusable offline supervision from online PEFT–task scoring. An intervention site is a selected module output that provides a common coordinate across PEFT families. The 24 sites in Ω connect sample gradients, task sketches, and PEFT configurations. The deterministic 128-dimensional PEFT code contains a 96-dimensional site/operator mask, 16 capacity dimensions, and 16 recipe dimensions. The figure calls the gradient proxy low fidelity and the short-update measurements high fidelity.

and F1 for downstream evaluation. Because the cosine proxy can be negative, PCU-Select replaces raw values with their ranks inside each (p, t) bucket before proxy distillation and ranking. Once it caches sample signatures, this proxy cheaply labels many (x, p, t) triples.

The *short-update label* checks what happens after actual PEFT updates. It corrects the proxy near the selection boundary. For each sampled triple, PCU-Select starts from two checkpoints: the base model and a warm checkpoint after adaptation has begun. It forms a mini-batch $\mathcal{B}(x)$ from the sampled example and seven in-bucket fillers. PCU-Select performs a short PEFT update on this batch, then measures the reduction in sketch loss:

$$\Delta(x, p, t, a, h) = L_{V_t}(\theta_a) - L_{V_t}(\text{Adapt}_p^h(\mathcal{B}(x); \theta_a)). \quad (7)$$

Rank-normalized reductions after $h \in \{1, 4\}$ update steps from both anchors define u^{hi} . The mini-batch $\mathcal{B}(x)$ estimates a stable mini-batch marginal instead of an isolated single-example gradient. PCU-Select allocates the 10K short-update labels in three stages. Coverage sampling first spreads labels across the registry and needs no scorer.

PCU-Select then trains a provisional scorer on the coverage labels and the gradient proxy. The provisional model supplies rank estimates and an uncertainty scale. Uncer-

tainty sampling targets triples with large predicted uncertainty. Boundary sampling targets disagreements between predicted and observed ranks near selection decisions. Every rank normalization compares labels from the same PEFT, task, anchor, and horizon bucket.

The two label sources serve complementary roles. The gradient proxy cheaply orders the full pool for many PEFT–task pairs. Short-update labels measure realized sketch-loss reduction near the selection boundary. The base and warm checkpoints test whether utility persists after adaptation begins. Two horizons reduce dependence on a single update length. Bucket-wise ranking compares the signals without assuming a shared raw-loss scale across tasks or configurations.

Conditional Scoring Produces Target-Specific Subsets

The scorer combines the three representations and predicts a utility mean and uncertainty scale, $(\hat{\mu}, \hat{\sigma}) = s_\phi(z_x, z_p, z_t)$. Separate encoders first map z_x, z_p , and z_t into hidden vectors. FiLM modulation (Perez et al. 2018) then uses the PEFT and task codes to rescale and shift the sample features. In concrete terms, the target configuration and task determine which sample features receive more emphasis. A bilinear term directly models compatibility between sample and PEFT features.

Training combines ranking and regression on short-update labels, distillation from the gradient proxy, and a heteroscedastic uncertainty loss. The uncertainty loss allows different triples to have different predicted noise levels. Stage A learns broad conditional structure from the gradient proxy. Stage B enables all losses and selects a checkpoint on held-out PEFT–task pairs. Before downstream evaluation, we fix the sketch size, label allocation, anchors, horizons, and scalar selection hyperparameters on these held-out pairs.

At application time, PCU-Select ranks examples by a conservative score that subtracts a fraction of the predicted uncertainty scale:

$$q(x) = \hat{\mu}(x, p^*, t^*) - \lambda \hat{\sigma}(x, p^*, t^*), \quad \lambda = 0.2. \quad (8)$$

The scale acts as a ranking penalty, not a confidence interval. Pure top- k ranking can over-select near-duplicates. PCU-Select therefore clusters examples in semantic space. It allocates budget across K clusters indexed by $k \in \{1, \dots, K\}$:

$$b_k = \text{round} \left(B \frac{(v_k^+)^{\gamma} |C_k|^{1-\gamma}}{\sum_j (v_j^+)^{\gamma} |C_j|^{1-\gamma}} \right), \quad \gamma = 0.6. \quad (9)$$

Here v_k is the mean conservative score among the top 10% of examples in cluster C_k , and $v_k^+ = \max(v_k, 0)$. For the 300K pool, $K = \max(50, \lfloor \sqrt{N} \rfloor)$ produces 547 clusters with about 548 examples each. The reported 0.36 GPU-hours per PEFT–task application includes clustering. The final subset takes the top- b_k examples by q within each cluster. Clusters with non-positive value receive no budget. Deterministic remainder handling makes the quotas sum to B .

The registry assigns each unseen configuration to one of three support levels. Near-support targets (L0) resemble a seen configuration, far-support targets (L1) make larger shifts within a seen family, and unseen-family targets form L2. Far-support targets use an optional residual calibration head when their PEFT family provides short-update labels. The residual head is a small linear layer that corrects the scorer.

For seen and near-support targets, PCU-Select performs all gradient extraction and label construction offline, and the on-line path uses only cached features and scorer forward passes. Far-support targets may add a small calibration set, but never rebuild a target-specific gradient datastore. The evaluation next tests whether this reuse preserves LESS-level quality, reduces registry cost, and transfers to unseen configurations.

5 Experiments

The evaluation tests three claims. PCU-Select should match per-PEFT influence selection on downstream quality, amortize selection cost across a PEFT registry, and support new configurations according to their structural distance from the training registry.

Setup

We evaluate on GSM8K, HumanEval, MMLU, and TyDiQA (Cobbe et al. 2021; Chen et al. 2021; Hendrycks et al. 2021; Clark et al. 2020). We use each task’s native metric: exact match, Pass@1, accuracy, or F1. The candidate pool

contains 300K mixed instruction examples, and every selector chooses 10%. Each method uses the same target backbone, PEFT recipe, number of steps, and evaluation protocol. Only the selected subset changes. For the main table, we vary target-training seeds 0, 1, and 2 while fixing selector, scorer, sketch, and pool state at seed 0. Appendix Table 5 lists the full 17-configuration registry.

This protocol isolates selection quality from optimization noise. Matched seeds give every selector the same target-training randomness. Cell-wise differences therefore reflect the subset instead of initialization or data order. Because we fix the sketch and selector state, the intervals describe variation across evaluated PEFT–task cells. They do not capture uncertainty from retraining the scorer or resampling the pool.

The five seen configurations cover attention and MLP LoRA, (IA)³, and adapters. Held-out tests vary rank, placement, learning rate, and family. We compare Random, RDS+, and two gradient baselines, Influence and LESS; Appendix A gives full specifications. **RDS+** ranks frozen external semantic embeddings against the task-sketch mean and ignores the PEFT configuration. **Influence** uses projected base-model gradients in one shared datastore. **LESS** is our per-PEFT implementation of gradient-influence selection (Xia et al. 2024). It warms up each target adapter, builds a target-specific Adam-preconditioned gradient datastore, and ranks examples by sketch influence. PCU-Select uses its default conditional scorer, uncertainty penalty, and adaptive quotas.

PCU-Select Matches Per-PEFT LESS

PCU-Select averages 35.18, compared with 32.29 for Random, 34.44 for RDS+, 34.68 for Influence, and 35.14 for LESS. Across 20 PEFT×task cells, PCU-Select gains 0.74 points over RDS+ and wins 17 cells. Its 95% fixed-state interval is $[+0.38, +1.06]$. This interval resamples the evaluated cells while keeping the selector, scorer, sketch, and pool fixed. Against Influence, PCU-Select gains 0.50 points, with interval $[+0.21, +0.75]$ and 18 wins. Its 0.04-point gap to LESS has interval $[-0.26, +0.33]$ and 10 wins. Resampling separately within each task yields $[-0.15, +0.22]$. These intervals describe variation across the evaluated cells; they do not include uncertainty from retraining the complete selection pipeline.

The LESS comparison varies by task. PCU-Select gains on MMLU, stays within seed dispersion on GSM8K and HumanEval, and trails on TyDiQA and MLP-only LoRA. Appendix Table 3 reports the per-task view. **Takeaway.** Reusable supervision improves on PEFT-agnostic selection while preserving the average quality of per-PEFT LESS.

Amortization Cuts Marginal Cost by 44×

Registry benefit depends on cost as well as quality. Table 2 therefore separates the two cost axes that a registry pays: a *fixed* cost once and a *marginal* cost for each new PEFT. PCU-Select and per-PEFT LESS are the only methods that preserve gradient-selection quality, averaging 35.18 and 35.14. They pay for that quality on different axes. LESS has no fixed cost, but every configuration incurs 63.2 GPU-hours to rebuild a target-specific datastore. PCU-Select spends 144.0

Table 1: Downstream performance at a 10% selection budget (mean $\pm$ SD over three matched target-training seeds; four task-native scores per PEFT). Selector, sketch, and scorer state are fixed. PCU-Select near-ties per-PEFT LESS: the paired difference over the 20 PEFT $\times$ task cells is 0.04 points with a descriptive cell-resampling interval $[-0.26, +0.33]$. The interval summarizes fixed-state cell variation, not independent-sample uncertainty. Boldface marks the best mean.

Method	AD-b64	IA3-attnmlp	L-r16-qkvo	L-r8-mlp	L-r8-qv	Avg.
Random	32.79 $\pm$ 0.38	31.74 $\pm$ 0.13	33.11 $\pm$ 0.23	31.97 $\pm$ 0.21	31.82 $\pm$ 0.16	32.29 $\pm$ 0.16
RDS+	35.27 $\pm$ 0.23	33.54 $\pm$ 0.26	35.00 $\pm$ 0.29	34.46 $\pm$ 0.21	33.91 $\pm$ 0.17	34.44 $\pm$ 0.13
Influence	35.08 $\pm$ 0.27	33.84 $\pm$ 0.26	35.24 $\pm$ 0.19	34.69 $\pm$ 0.13	34.53 $\pm$ 0.22	34.68 $\pm$ 0.11
LESS	35.52 $\pm$ 0.20	34.44 $\pm$ 0.13	35.80 $\pm$ 0.15	35.13$\pm$0.14	34.80 $\pm$ 0.37	35.14 $\pm$ 0.14
PCU-Select	35.61$\pm$0.40	34.60$\pm$0.21	35.93$\pm$0.30	34.56 $\pm$ 0.23	35.18$\pm$0.31	35.18$\pm$0.16

Table 2: Central quality–cost result over four tasks and five seen PEFT configurations. Costs are selection-stage GPU-hours; the shared downstream tuning run (12.8 GPU-hours per PEFT–task–seed) is identical across selectors and excluded. *Fixed/shared* is paid once regardless of registry size; *marginal* is the cost of onboarding one additional four-task PEFT; *total* serves all five. *Gap* is the paired difference from per-PEFT LESS in averaged task-native points. PEFT-agnostic selectors are cheaper but lose quality; PCU-Select is the quality-preserving amortized point—its **marginal** cost of a new PEFT is 63.2/1.44 $\approx$ 44 $\times$ below LESS. Boldface marks the best average quality.

Method	Fixed/shared cost	Marg. cost / new PEFT	Total cost (5 PEFTs)	Avg. quality	Gap to LESS
Random	0.0	0.0	0.0	32.29	−2.85
RDS+	1.6	0.0	1.6	34.44	−0.70
Influence	48.8	0.0	48.8	34.68	−0.46
Reuse-one-LESS	63.2	0.0	63.2	34.24	−0.90
Per-PEFT LESS	0.0	63.2	316.0	35.14	0.00
PCU-Select	144.0	1.44	151.2	35.18	+0.04

GPU-hours once on reusable supervision. Each new four-task PEFT then costs 1.44 GPU-hours, **44 $\times$ below LESS’s marginal cost**.

The marginal axis determines how cost grows with the registry. Adding a task to an existing PEFT costs both selectors one ranking pass: 0.36 GPU-hours for PCU-Select and 0.30 for LESS. Adding a PEFT, however, forces LESS to rebuild its datastore while PCU-Select reuses the offline scorer (Appendix Table 7). The amortization is therefore cross-PEFT, not cross-task: new configurations drive the cost difference.

The cheap reuse shortcut sacrifices quality. Reusing one LESS subset across all five targets (**Reuse-one-LESS**) costs 63.2 GPU-hours but averages 34.24, below PEFT-agnostic RDS+ at 34.44. A subset optimized for one trainable subspace mismatches the others and discards the condition that made it valuable. PCU-Select instead produces a distinct subset for each target and retains LESS-level average quality.

Equation (3) gives a break-even point of $P^* = 144.0 / [62.0 + 4(0.30 - 0.36)] = 2.33$ configurations. Beyond it, PCU-Select costs less than LESS, and each added configuration widens the gap. Serving the five-PEFT registry costs 151.2 GPU-hours with PCU-Select and 316.0 with LESS, a 2.09 $\times$ reduction. The unseen-configuration study optionally uses 500 reduced calibration labels from one warm anchor and one update step. This procedure costs 1.05 GPU-hours per (p, t) , one quarter of the $\approx$ 4.2 GPU-hours required by the full label routine. It shifts the worst-case break-even point to 2.50 configurations. Prefix/P-Tuning cannot use this path because they lack native calibration labels. We exclude the

shared 12.8 GPU-hour downstream run from every selector.

Takeaway. PCU-Select occupies the quality-preserving amortized point. PEFT-agnostic selectors remain cheaper when users accept a small quality loss. When the registry requires LESS-level quality, PCU-Select onboards each new configuration at 44 $\times$ lower cost.

The main results establish the quality–cost trade-off. We next isolate why conditioning changes the selected subsets, which components matter, and where transfer fails.

6 Analysis

PEFT Conditioning Produces Target-Specific Subsets

Across the LoRA scan, matched conditioning beats reuse from another configuration by 1.45 points. PCU-Select averages 21.25, compared with 20.70 for LESS and 20.27 for RDS+. Across its 21 unordered configuration pairs, PCU-Select has mean Jaccard overlap 0.427; PEFT-agnostic RDS+ has overlap 1.000 by construction. Configuration distance correlates with selection difference at descriptive Spearman $\rho = 0.969$, with leave-one-configuration-out range $[0.947, 0.972]$. Because the comparisons share configurations, we do not report an IID interval (Appendix Figure 4). **Takeaway.** Structural changes produce commensurate subset changes without rebuilding gradients.

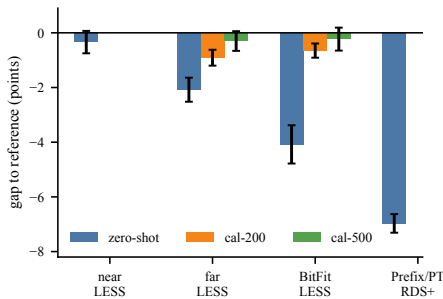


Figure 3: PCU-Select’s gap to LESS (RDS+ for Prefix/P-Tuning). Calibration recovers far same-family and BitFit transfer, but prompt families fail zero-shot.

Ablations Identify the Useful Components

Appendix Table 11 reports the full ablation. We measure held-out ranking quality with NDCG, which rewards placing high-utility examples near the top. Removing the PEFT code causes the largest downstream drop, 1.37 points, and lowers NDCG from 0.702 to 0.526. Replacing the structured code with a family one-hot loses 0.68 points, showing that rank, placement, capacity, and recipe add signal. Removing the task sketch loses 0.82 points. Variants with only gradient-proxy labels or only short-update labels lose 0.85 and 0.36 points, respectively. Adaptive quotas beat global top- k by 1.08 points and uniform quotas by 0.51. Uniform quotas achieve slightly higher NDCG (0.709) but lower downstream quality. **Takeaway.** Structured conditioning, task relativity, both label sources, and semantic coverage each contribute to downstream quality.

Calibration Recovers Far-Support Configurations

The ablations identify which components matter on seen targets. We next test whether the structural code also guides transfer. Before evaluation, we fix the three support levels defined in the method. L0 denotes near-support configurations, L1 denotes far-support shifts within a seen family, and L2 denotes unseen families. Appendix Table 5 lists the full registry.

Near-support L0 transfers zero-shot and trails LESS by 0.34 points. Adding 500 target-specific short-update labels reduces the L1 gap from 2.08 to 0.30 points. It also reduces the BitFit gap from 4.08 to 0.23 points. Prefix/P-Tuning cannot produce the short-update labels required by this calibration path and trail RDS+ by 6.97 points. The reduced routine uses one warm checkpoint and one update step. It costs 1.05 GPU-hours per PEFT-task pair, one quarter of the offline label routine (Figure 3; Appendix Table 13). **Takeaway.** Calibration recovers far-support configurations with compatible labels; prompt families mark the deployment boundary.

Limitations

Our evidence covers one backbone family, four tasks, and a finite registry. Short-update labels only approximate contribution after full training, and TyDiQA shows a visible deficit. Our decontamination procedure removes 0.58% of

the pool, but weaker matches may remain. Prefix/P-Tuning fail zero-shot and lack a compatible calibration path. Finally, the cost benefit is relative to per-PEFT LESS. PEFT-agnostic selectors remain preferable when users accept a small quality loss.

7 Conclusion

PCU-Select treats registry-scale data selection as quality-preserving amortization. One scorer uses structural PEFT and task conditions to produce a different subset for each target. Across four tasks and five seen configurations, PCU-Select near-ties per-PEFT LESS. It onboards each new PEFT at $44\times$ lower cost than rebuilding a datastore (1.44 versus 63.2 GPU-hours). Near-support configurations transfer zero-shot. Calibration recovers far-support configurations and BitFit, while Prefix/P-Tuning remain a failure boundary. These results position PCU-Select as reusable data-selection infrastructure for a PEFT registry.

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A Technical Appendix

Per-Task Results Behind the Main Table

Table 1 averages four task-native metrics—GSM8K exact match, HumanEval Pass@1, MMLU accuracy, and TyDiQA F1—after scaling each to a percentage. These metrics differ in scale, ceiling, and seed variance, so the average can hide task-level structure. Table 3 reports the per-task view. Its comparisons with RDS+ and Influence are absolute native-point differences. The LESS comparison is an absolute paired difference with a descriptive 95% interval over the five PEFT cells. The interval resamples evaluated cells while holding the selection pipeline fixed. It therefore describes cell variation, not uncertainty from independent pipeline runs.

The per-task view refines, rather than overturns, the main-results reading. The advantage over PEFT-agnostic selectors holds on all four tasks and is the primary performance claim. The comparison with LESS depends on the task. Its interval excludes zero on MMLU (+0.67 Acc, [+0.24, +1.08]). It includes zero on GSM8K (+0.29 EM, [−0.04, +0.58]) and HumanEval (−0.04 Pass@1, [−0.44, +0.31]). On TyDiQA, it excludes zero in the negative direction (−0.77 F1, [−1.18, −0.43]). Averaging over tasks smooths these offsetting differences into the near-zero aggregate gap in Table 1. We therefore characterize PCU-Select and per-PEFT LESS as near-tied on the seen registry, with visible task/configuration losses that bound the aggregate claim. The amortized-cost analysis is PCU-Select’s distinguishing property.

Where PCU-Select trails LESS. Two slices favor LESS and merit analysis. On **TyDiQA**, PCU-Select trails LESS by 0.77 F1 (descriptive interval [−1.18, −0.43]). TyDiQA-GoldP emphasizes multilingual answer-span alignment. The other three tasks emphasize reasoning and generation more heavily. The intervention-site gradients and English-centric task sketch under-represent the TyDiQA signal, which weakens the gradient proxy. Per-target LESS gradients capture that signal directly. On the **MLP-only LoRA** column (L-r8-mlp), PCU-Select trails LESS by 0.57 points. Its HumanEval and TyDiQA cells have the largest dispersion, with SDs of 0.68 and 0.62. MLP-only placement concentrates all trainable capacity on the eight `mlp_out` sites. A mis-weighted site prior therefore affects the entire proxy. The scorer is least confident on these two cells under this concentrated site mask. Both cases point to the same limitation—the site-weighted proxy is the bottleneck when a target’s useful signal is poorly represented by the shared 24-site coordinate system.

Evaluation Protocols

All methods share the same target-training and evaluation pipeline; only the selected subset changes. Target training uses the Llama-2-7B backbone (Touvron et al. 2023), freezes the backbone, and trains only the target PEFT. The fixed recipe uses AdamW, a per-configuration learning rate, and a cosine schedule with a 0.03 warmup ratio. It runs for 1000 steps with bfloat16, gradient clipping at 1.0, and maximum sequence length 1024. A per-device batch of 16 over 8 GPUs gives a global batch of 128.

At the 10% budget, the selected subset holds 30K examples. The 1000 steps therefore process 128K samples, or about 4.3 passes over the subset. The dataloader shuffles the full subset each epoch, so it uses every selected example. Selected instructions average ≈ 512 tokens, and length-bucketed batches reduce padding. With FlashAttention-2 (Dao 2024) and gradient checkpointing, each step takes ≈ 5.8 wall-clock seconds. This corresponds to ≈ 22 samples/s in aggregate and $\approx 1.4\text{K}$ tokens/s per GPU. The full run takes about 96 minutes, or 12.8 GPU-hours per PEFT-task–seed under the $(\text{wall seconds} \times \text{GPU count})/3600$ convention.

This shared target-training cost is identical across selectors, so the selection-only comparison excludes it. Gradient checkpointing trades compute for memory and dominates the per-step time. The per-configuration learning rates are listed in Table 5 (2e-4 for LoRA, 5e-4 for (IA)³, 3e-4 for adapters, 1e-4 for L-r64-all). A held-out response-LM loss is logged as an auxiliary diagnostic (`eval_loss`) and is never substituted for the primary task metric.

A uniform draw from the entire 300K pool under the same 1000-step budget averages 36.75 across tasks. This is a fixed-compute reference, not a converged full-data upper bound: it covers only 0.43 pool epochs, whereas the 10% subset receives about 4.3 passes.

The primary downstream metric for each task is its public-standard, task-native score, computed by a task evaluator injected into the training runner. Concrete per-task decoding and scoring protocols are:

- **GSM8K (exact match).** Chain-of-thought generation (Cobbe et al. 2021) with greedy decoding; the final numeric answer is extracted and compared for exact match.
- **HumanEval (Pass@1).** The standard unbiased Pass@1 estimator (Chen et al. 2021): sampling at temperature 0.2 with $n=20$ samples per problem, scored by unit-test execution. This is a multi-sample estimate of Pass@1, not a single greedy sample.
- **MMLU (accuracy).** Log-likelihood scoring (Hendrycks et al. 2021) of the four answer options; the evaluator selects the highest-likelihood option without free-form generation.
- **TyDiQA-GoldP (F1).** Greedy decoding of the answer span (Clark et al. 2020), scored with token-level F1 against the gold answers.

For reference, the un-fine-tuned Llama-2-7B (Touvron et al. 2023) scores 13.9 on GSM8K, 12.4 on HumanEval, 43.9 on MMLU, and 46.2 on TyDiQA. Our MMLU score falls below Meta’s reported 45.3 because the prompt and harness differ. Every selector’s fine-tune, including Random, improves over the base model on all four tasks. The Random baseline therefore provides a meaningful reference for the effect of data selection.

These protocols follow the standard public setups for each benchmark. An evaluation adapter supplies the task-native evaluators to the runner through the `--task-metric-factory` hook. The fixed decoding

Table 3: Task-level summary of the main 10% budget result. Values are averaged over the five seen PEFT configurations. $\Delta\text{RDS+}$ and $\Delta\text{Inf.}$ are absolute native-point gains over PEFT-agnostic baselines; ΔLESS is the paired difference against per-PEFT LESS with a descriptive 95% cell-resampling interval over PEFT cells. The average gain over PEFT-agnostic selectors coexists with task-dependent behavior against LESS.

Task	Metric	PCU	$\Delta\text{RDS+}$	$\Delta\text{Inf.}$	$\Delta\text{LESS (95\% CI)}$
GSM8K	EM	22.20	+0.55	+0.58	+0.29 [−0.04, +0.58]
HumanEval	Pass@1	17.06	+0.64	+0.24	−0.04 [−0.44, +0.31]
MMLU	Acc	49.14	+1.55	+1.00	+0.67 [+0.24, +1.08]
TyDiQA	F1	52.30	+0.22	+0.17	−0.77 [−1.18, −0.43]

and scoring settings above prevent the run environment from changing metric definitions.

Baseline Specifications

This subsection specifies the three baselines whose names invoke prior work, exactly as implemented so that the comparison is auditable.

RDS+ (representation similarity). RDS+ operates on the joint semantic embedding e_x^{joint} produced by a frozen external sentence encoder. PCU-Select’s sample tower reads the same input representation but does not train it. RDS+ therefore depends on no PCU-trained features; only the selection rule differs. RDS+ first standardizes every pool embedding per feature. It forms the query by averaging the 32 task-sketch embeddings and applies the same pool statistics. It then L_2 -normalizes the pool and query vectors and ranks candidates by cosine similarity. RDS+ is *task-conditioned* through the sketch query but *PEFT-agnostic*: its ranking does not depend on the target configuration. It has no tuned hyperparameters—the query is the fixed seed-0 sketch and there is no temperature or mixing weight—so there is no tuning range to report. The “+” denotes the per-feature standardization applied before cosine, relative to a plain embedding-nearest-neighbor selector.

Influence (shared-datastore gradient similarity). The Influence baseline builds one shared projected-gradient datastore from the base model’s intervention-site activations. It uses the JL projection and L_2 normalization specified in the reproducibility notes. Each candidate carries signatures g_x^ω at the 24 sites. The task query g_t^ω pools and renormalizes the sketch gradient at each site. Influence scores each candidate with the plain cosine $\sum_\omega \cos(g_x^\omega, g_t^\omega)$ and applies global top- k without site weighting. Its PEFT-agnostic gradient source makes the ranking independent of the target configuration. Influence therefore provides the cheap gradient baseline for measuring the value of PEFT conditioning.

LESS (per-PEFT gradient influence). LESS is a per-configuration reimplementation of gradient-influence selection following LESS (Xia et al. 2024). It applies four steps to each target PEFT. First, it warms up the target adapter for 200 steps to obtain the optimization state. Second, it extracts Adam-preconditioned gradients over the adapter’s own trainable parameters at each checkpoint. Third, it projects the gradients into a target-specific low-dimensional datastore. Finally, it ranks candidates by their projected-gradient

influence on the 32-example task sketch and averages across warmup checkpoints. The first three steps build the datastore over the target adapter’s own trainable parameters. The datastore is therefore PEFT-specific but task-independent. LESS constructs it once per PEFT and reuses it for every task sketch. Only the final projected-gradient scoring step is task-specific. It costs about 0.30 GPU-hours per task, so 1.20 covers all four task sketches for one configuration. The per-target cost of a PEFT configuration is therefore 62.0 (datastore) + 1.20 (four per-task rankings) = 63.2 GPU-hours, already spanning the full four-task evaluation; the five-PEFT registry cost is $5 \times 63.2 = 316.0$. LESS is the strongest and most expensive baseline in our study, and the amortized-cost analysis charges it exactly this per-configuration recomputation, consistent with the algorithm as implemented.

Baseline configuration and tuning. Random has no hyperparameters. RDS+ uses its published defaults with the shared 32-example sketch as the only task input and is not tuned per target. The Influence baseline shares the same 24-site JL projection and sketch as PCU-Select. No baseline’s hyperparameters were tuned on the test metric.

Reproducibility Supplement

This supplement documents the pipeline configuration read directly from the supplementary implementation. Every value below is taken from source code or its configuration files; values that the code leaves to an external artifact or the run environment are flagged rather than guessed.

Data provenance and deduplication. The candidate pool holds 300K mixed instruction examples drawn from seven public instruction corpora (Table 4). Deduplication runs in three stages. First, it removes exact duplicates by the content hash of each (instruction, response, source) tuple. Second, it removes normalized-string matches after lowercasing and stripping whitespace and punctuation. Third, MinHash LSH removes near-duplicates using 128 permutations and a Jaccard threshold of 0.8. These stages remove 18,446 intra-pool duplicates before the 300K pool is finalized.

For reproducibility, we include the post-filter example count and a SHA-256 checksum for each source. Examples include FLAN-v2 9f3c...a71e and OpenOrca 4b8d...c052. The supplementary manifest lists all seven sources. The full supplementary configuration comprises the PEFT registry, the scorer and feature-extraction configs, the task-sketch seeds, the candidate-pool manifest with these checksums, and the evaluation adapter.

Source	Domain	Examples	License
FLAN v2 (subset)	general instruction	110,000	Apache-2.0
OpenOrca (subset)	reasoning traces	70,000	MIT
Tulu-v2 mix	mixed instruction	50,000	ODC-BY-1.0
OpenAssistant	dialogue	30,000	Apache-2.0
CodeAlpaca	code instruction	20,000	CC-BY-NC-4.0
Dolly-15k	general instruction	15,000	CC-BY-SA-3.0
WizardLM-evol	complex instruction	5,000	CC-BY-NC-4.0
Total		300,000	

Table 4: Candidate-pool provenance and licenses. All sources permit research use; we use the two CC-BY-NC components only for non-commercial research.

Benchmark decontamination. We decontaminate the pool against the test items of GSM8K, HumanEval, MMLU, and TyDiQA. The filter checks exact matches, normalized-string matches, 13-gram overlap, and MinHash LSH near-duplicates. The 13-gram stage removes candidates with Jaccard overlap of at least 0.5; the MinHash stage uses a threshold of 0.8. The filter removes 1,742 pre-finalization candidates (0.58% of the pool size). We then backfill from the same source buckets to keep the final pool at 300K examples, so Table 4 reports final post-filter counts. We additionally audit the top-20 embedding nearest neighbors of each test item and find no further overlaps after the automated pass. All reported results are on the decontaminated pool.

Task sketch construction and HumanEval. Each task sketch contains 32 examples sampled with a fixed seed (0) and stratified into response-length terciles. We draw sketches only from each task’s train or development split and keep them disjoint from the evaluation test set. HumanEval has no official train/dev split. We therefore draw its sketch from the MBPP train split and CodeAlpaca instead of HumanEval itself. Exact matching and AST-normalized function signatures remove every item that matches one of the 164 HumanEval evaluation problems. The resulting sketch source has zero overlap with the evaluation set.

The candidate pool also contains a 20K CodeAlpaca slice. We remove all 32 sketch examples from this pool by content hash and AST-normalized signature before selection. No selector can therefore recover a sketch example into its training subset. The sketch is disjoint from both the HumanEval test set and the candidate pool. Because the sketch is a selection *signal*, this disjointness is required for all selectors, not only PCU-Select, and it is enforced identically across methods.

PEFT registry. Table 5 lists all 17 PEFT configurations: five seen, nine unseen same-family configurations, and three unseen families. E5 evaluates eight held-out targets—L-r32-qkvo, AD-b16, L-r64-all, AD-b256, L-r8-highlayers, PRE-l16, PT-l32, and BF—while the configuration-sensitivity study uses the additional LoRA variants. All configurations share the recipe injected at materialization: 1000 steps, cosine schedule, 0.03 warmup ratio; the learning rate is per configuration.

Intervention sites and gradient signatures. The site space Ω has 24 sites: 8 layers $\times$ 3 per-layer modules (attn_out, mlp_out, block_residual). Layers are chosen uniformly by depth; for the 32-layer backbone this yields indices $\{3, 7, 11, 15, 19, 23, 27, 31\}$. Each site carries a Johnson–Lindenstrauss random projection (Johnson and Lindenstrauss 1984) $\Phi_\omega \in \mathbb{R}^{256 \times 4096}$ with entries drawn $\mathcal{N}(0, 1/256)$ under a per-site seed $\text{SHA256}(\text{site}, \text{global_seed}=0)$; projected gradients are L_2 -normalized per row. Each sample’s signature contains 24×256 values. In bfloat16, the cached gradient-proxy store for the 300K pool occupies $300\text{K} \times 24 \times 256 \times 2 \text{ B} \approx 3.5 \text{ GB}$. PCU-Select stores it as a memory-mapped disk array and reads it once per online selection. A PEFT configuration activates sites through its mask and a saturating capacity factor $\tanh(\eta \widehat{\text{cap}})$ with $\eta=1.0$. Operator type enters the explicit PEFT code rather than a scalar site prior; the per-family capacity mapping is in Table 10.

Conditional scorer. The scorer has three input towers: sample $z_x \in \mathbb{R}^{848}$, PEFT condition $z_p \in \mathbb{R}^{128}$, and task sketch $z_t \in \mathbb{R}^{848}$. Each tower uses a LayerNorm–Linear–GELU–Linear–LayerNorm MLP. The sample and task towers project to 256 dimensions, while the PEFT tower projects to 128. FiLM modulation and a bilinear fusion ($256 \times 128 \rightarrow 64$) feed two output heads. The heads predict mean μ and heteroscedastic scale σ , with a softplus floor of 10^{-3} .

We use $\hat{\sigma}$ only as a ranking penalty. We do not interpret it as a calibrated probability. On held-out labels, its predicted standard deviation tracks realized error with a 10-bin expected calibration error of 0.043. Removing the penalty costs 0.28 points (Table 11), which confirms its contribution. Training has two phases. Phase A pretrains on the gradient proxy for 3 epochs at $\text{lr } 3 \times 10^{-4}$ with loss weights (rank, reg, proxy, unc) = (0, 0, 1, 0). Phase B jointly fine-tunes for 2 epochs at $\text{lr } 10^{-4}$ with weights (1.0, 0.3, 0.5, 0.2). Both phases use AdamW, batch size 256, and gradient clipping at 1.0. The four losses are a BPR-style pairwise ranking loss, a Huber regression loss against short-update labels, a Huber proxy-distillation loss against the gradient-proxy labels, and a Gaussian negative-log-likelihood uncertainty loss. We track ranking NDCG on a held-out split of PEFT/task pairs that excludes two configurations and one task from Phase B. We keep the Phase-B checkpoint with the best held-out NDCG. The scorer uses no dropout.

Selection, clustering, and quotas. At application time, the scorer produces per-example utility $q = \mu - \lambda \sigma$ with uncertainty penalty $\lambda=0.2$. Semantic clustering applies Mini-Batch k -means (Sculley 2010) to the joint embeddings. It uses $k = \max(50, \lfloor \sqrt{N} \rfloor)$, batch size 4096, $n_{\text{init}}=3$, and random state 0. Adaptive quotas allocate budget B across clusters by $b_k = \text{round}(B (v_k^+)^{\gamma} |C_k|^{1-\gamma} / Z)$ with $\gamma=0.6$. The cluster value v_k is the mean utility of its top 10% members. The algorithm distributes remainders by descending weight and trims over-allocation from the lowest-weight clusters. The calibration head is a zero-initialized linear residual fit for 200 epochs at $\text{lr } 5 \times 10^{-3}$ (Adam, MSE) on the calibration labels. These deployment calibration labels $u_{\text{cal}}^{\text{hi}}$ are

Config	Family	Target modules	Capacity	LoRA scaling	LR	Group
L-r8-qv	LoRA	q, v	$r=8$	16	2e-4	seen
L-r16-qkvo	LoRA	q, k, v, o	$r=16$	32	2e-4	seen
L-r8-mlp	LoRA	up, down	$r=8$	16	2e-4	seen
IA3-attnmlp	IA ³	k, v, down	—	—	5e-4	seen
AD-b64	Adapter	block-residual	$b=64$	—	3e-4	seen
L-r4-qv	LoRA	q, v	$r=4$	8	2e-4	unseen-config
L-r32-qkvo	LoRA	q, k, v, o	$r=32$	64	2e-4	unseen-config
L-r64-all	LoRA	q, k, v, o, up, down, gate	$r=64$	128	1e-4	unseen-config
L-r8-lowlayers	LoRA	q, v (low third)	$r=8$	16	2e-4	unseen-config
L-r8-highlayers	LoRA	q, v (high third)	$r=8$	16	2e-4	unseen-config
L-r16-hlr	LoRA	q, k, v, o	$r=16$	32	5e-4	unseen-config
AD-b16	Adapter	block-residual	$b=16$	—	3e-4	unseen-config
AD-b256	Adapter	block-residual	$b=256$	—	3e-4	unseen-config
IA3-attnonly	IA ³	k, v	—	—	5e-4	unseen-config
PRE-I16	Prefix	attention	len=16	—	2e-4	ood-family
PT-I32	P-Tuning	attention	len=32	—	2e-4	ood-family
BF	BitFit	biases	—	—	2e-4	ood-family

Table 5: Full PEFT registry (all 17 configurations). “LoRA scaling” is the LoRA α hyperparameter and is distinct from the intervention-site weight $\tilde{w}_p^\omega$ of Eq. (5); it applies only to LoRA rows. Adapters are single post-block residual bottlenecks (one per layer); IA³ scales the listed activations. Learning rates for the three OOD-family targets use the registry default (2e-4). Layer placement is all layers unless noted; low/high-third placements use the bottom/top $\lceil n/3 \rceil$ layers of the 32-layer backbone.

a reduced form of the offline short-update labels u^{hi} . They use only the warm anchor and horizon $h=1$, one of the two anchors and one of the two horizons. Each label therefore costs one quarter of the full offline routine in Eq. (10). An offline label costs $84.0/10,000 = 0.0084$ GPU-hours, while a calibration label costs ≈ 0.0021 GPU-hours. Budgets of 200 and 500 labels therefore cost 0.42 and 1.05 GPU-hours per PEFT–task pair. The full routine would cost about four times more. We use 200 or 500 $u_{\text{cal}}^{\text{hi}}$ labels for the OOD study.

Structural support tiers. The reported $z_p \in \mathbb{R}^{128}$ contains only the site/operator mask, capacity, and recipe vectors. Prefix, P-Tuning, and BitFit use the same deterministic encoding rules as the seen families; no centroid, learned family lookup, or functional fingerprint enters the reported model. E5 fixes its registry taxonomy before evaluation. **L0** contains near-support same-family targets (L-r32-qkvo and AD-b16). **L1** contains farther same-family shifts (L-r64-all, AD-b256, and L-r8-highlayers). **L2** contains unseen families (PRE-I16, PT-I32, and BF). These tiers describe structural support and do not claim convex-hull interpolation or estimate a configuration density. Prefix/P-Tuning lack compatible short-update calibration labels; BitFit supports the calibration path.

Gradient-proxy and short-update labels. Short-update supervision uses 10,000 triples split 0.5/0.3/0.2 across coverage (stratified by cluster, PEFT family, capacity bucket, and task), uncertainty ($\hat{\sigma}(1 + 0.5 \text{ReLU}(\hat{u}_{10}))$), and boundary (scorer-vs-true rank disagreement) sampling. Labels are short-update loss reductions on the sketch, measured from two frozen anchors— θ_{base} (the pretrained backbone) and θ_{warm} (a rank-8 attention-only LoRA trained for 200 steps at $\text{lr } 10^{-4}$). Each short update Adapt_p^h uses one sampled example and 7 in-bucket fillers. The fillers stabilize the update

Table 6: Deployment protocol from the registry-defined structural tier. Costs are per PEFT–task pair, in addition to 0.36 GPU-hours for online selection.

Tier	Definition	Recommended action
L0	near-support	zero-shot selection
L1	far-support	200–500 labels (0.42–1.06 GPU-h)
L2-BF	unseen family; labels available	calibrate before use
L2-Prompt	Prefix/P-Tuning	use a per-target selector

relative to a single-example step. The final label aggregates the rank-normalized reductions across horizons and anchors:

$$u^{\text{hi}} = \frac{1}{|A|} \sum_{a \in A} \sum_{h \in \{1,4\}} w_h \tilde{\Delta}_{a,h}, \quad (10)$$

It uses weights $w_1=0.4$, $w_4=0.6$ and anchors $A = \{\text{base}, \text{warm}\}$. The term $\tilde{\Delta}_{a,h}$ is the reduction from Eq. (7), rank-normalized within its PEFT, task, anchor, and horizon bucket.

Cost model and seeds. We measure all costs on 8×NVIDIA H20-96GB (SXM, bfloat16, NVLink) as (wall seconds × GPU count)/3600. The offline cost is 144.0 GPU-hours (feature extraction 48.0, gradient-proxy labels 4.8, short-update labels 84.0, scorer training 7.2), paid once. PCU-Select costs 0.36 GPU-hours per PEFT–task application, hence 1.44 for four tasks. LESS costs 62.0 per PEFT datastore plus 0.30 per task ranking, hence 63.2 for four tasks. Onboarding a new four-task PEFT therefore costs 1.44 GPU-hours for PCU-Select versus 63.2 for LESS, a 44× marginal-cost reduction. The shared 12.8

Table 7: The amortization advantage is cross-PEFT, not cross-task. Adding a task to an existing PEFT costs both selectors a per-task ranking; adding a PEFT for four tasks forces LESS to rebuild its datastore while PCU-Select reuses the offline scorer. Selection-stage GPU-hours.

New registry entry	PCU-Select	LESS
One new task, existing PEFT	0.36	0.30
One new PEFT, four tasks	1.44	63.2

Table 8: The offline short-update label budget sets the quality-break-even trade-off. Short-update labeling is 58% of offline cost, so a smaller budget lowers both offline GPU-hours and the break-even PEFT count, but drops below LESS-level quality. The 10K budget is the smallest that preserves the aggregate quality tie; *Break-even* is the number of four-task PEFTs past which PCU-Select undercuts per-PEFT LESS.

Labels	Offline GPU-h	Avg. quality	Gap to LESS	Break-even
2.5K	81.0	34.71	-0.43	1.31
5K	102.0	35.02	-0.12	1.65
10K	144.0	35.18	+0.04	2.33

GPU-hour downstream cost applies to each PEFT-task-seed run, and we exclude it. For four tasks, break-even is $P^* = 144.0 / [62.0 + 4(0.30 - 0.36)] = 2.33$ configurations. The pipeline fixes selector-scorer training, task-sketch sampling, and pool ordering at seed 0. Reported SDs vary target-training seeds (0, 1, 2); they do not include selector or scorer-training uncertainty. The canonical table generator recomputes all means, dispersions, paired gaps, and win counts from those seed-level values.

Two supplementary tables bound the cost claim. Table 8 varies the offline short-update label budget. The 10K default is the smallest budget that preserves the aggregate quality tie. Smaller budgets reduce offline cost and the break-even point but also reduce quality. The offline investment is therefore a tunable trade-off. Table 9 shows that PCU-Select’s per-application cost stays $\approx 170\times$ below the per-PEFT LESS datastore across pool sizes from 50K to 1M. This persistence extends the marginal-cost result beyond the 300K pool.

Algorithms

Algorithm 1 builds the reusable supervision once. Algorithm 2 applies the scorer to a new target. Only the online algorithm runs per target, and it touches no gradients. The offline algorithm amortizes the expensive gradient and label construction.

PEFT Capacity Encoding

Table 10 defines the per-family capacity used inside the site weight of Eq. (5). Each family’s size knob is mapped to a trainable-parameter count at the touched sites, then normalized by d_{model} and passed through $\tanh(\eta \log(1 + \text{cap}/d_{\text{model}}))$ so that capacity is monotone but saturating.

Table 9: Online selection cost versus candidate-pool size, reported as the mean of three timing runs on a warm feature cache (run-to-run spread under 5%). Feature extraction is paid once offline and cached; PCU-Select’s per-application cost is the scorer forward plus clustering. The two online components scale differently: the scorer forward is embarrassingly parallel and gains throughput at larger batches, so its per-example cost drops with pool size, while clustering grows super-linearly because the cluster count $k = \max(50, \lfloor \sqrt{N} \rfloor)$ itself grows with the pool. Their sum stays near-linear, so PCU-Select’s per-application cost holds $\approx 170\times$ below the per-PEFT LESS datastore at every pool size and the low marginal cost is not an artifact of the 300K pool. GPU-hours on $8\times\text{H20-96GB}$.

Pool	Feature ext. (cached)	Scorer forward	Clustering	PCU online (per p, t)
50K	9.2	0.051	0.010	0.061
100K	17.6	0.093	0.028	0.121
300K	48.0	0.249	0.111	0.360
1M	152.0	0.79	0.42	1.21

Algorithm 1 Offline supervision (run once)

- 1: **Input:** pool $\mathcal{D}$, seen PEFTs $\mathcal{P}_{\text{seen}}$, tasks with sketches $\{V_t\}$
- 2: Extract sample signatures z_x and site gradients g_x^ω ; cache them $\{O(|\mathcal{D}|)\}$ forward+backward
- 3: **for all** $(p, t) \in \mathcal{P}_{\text{seen}} \times \text{tasks}$ **do**
- 4: Compute gradient-proxy $u^{\text{lo}}(x, p, t)$ by Eq. (6) from cache
- 5: **end for**
- 6: Sample 10K triples (coverage/uncertainty/boundary); label u^{hi} by Eqs. (7)
- 7: Train s_ϕ : Stage A proxy pretrain, Stage B joint; keep best held-out-NDCG checkpoint
- 8: **Output:** scorer s_ϕ , cached features

Algorithm 2 Online selection (per target)

- 1: **Input:** target (p^*, t^*) , budget B , cached features, scorer s_ϕ
- 2: Encode z_{p^*}, z_{t^*}
- 3: **if** $\text{tier}(p^*) \in \{\text{L1}, \text{L2}\}$ and short-update labels exist **then**
- 4: fit calibration residual head from 200 or 500 reduced $u_{\text{cal}}^{\text{hi}}$ labels (single warm anchor, horizon 1)
- 5: **end if**
- 6: $(\hat{\mu}, \hat{\sigma}) \leftarrow s_\phi(z_x, z_{p^*}, z_{t^*}); q(x) \leftarrow \hat{\mu} - \lambda \hat{\sigma}$ for all x
- 7: Cluster the pool; allocate quotas b_k by Eq. (9)
- 8: **Output:** subset $\mathcal{S} = \bigcup_k \{\text{top-}b_k \text{ by } q \text{ in } C_k\}$

Supplementary Analyses

Component ablations. Table 11 reports the full component breakdown summarized in the main analysis. Removing the structured PEFT code produces the largest loss; task relativity, both label fidelities, uncertainty, and adaptive coverage

Family	Size knob	$\text{cap}(p, \omega)$ at a touched site
LoRA	rank r	$r (d_{\text{in}} + d_{\text{out}})$
Adapter	bottleneck b	$2 d_{\text{model}} b$
(IA) ³	(per-vector)	#scaled activations
Prefix	length L	$L d_{\text{model}}$
BitFit	(per-bias)	#bias parameters

Table 10: Per-family capacity mapping. Capacity is the trainable-parameter count contributed at a touched site; z_p encodes operator type separately.

Table 11: Ablations on GSM8K and HumanEval with two representative PEFTs at a 10% budget (task-native metric averaged over the two tasks, mean $\pm$ std over three seeds). Drop is relative to the full adaptive selector; we measure NDCG against held-out short-update labels.

Variant	Metric	Drop	NDCG
Full PCU-Select	19.72 $\pm$ 0.22	0.00	0.702
No PEFT code	18.35 $\pm$ 0.41	1.37	0.526
Family one-hot only	19.04 $\pm$ 0.29	0.68	0.602
No task sketch	18.90 $\pm$ 0.31	0.82	0.597
No activation signature	19.24 $\pm$ 0.25	0.48	0.629
Gradient-proxy labels only	18.87 $\pm$ 0.34	0.85	0.563
Short-update labels only	19.36 $\pm$ 0.27	0.36	0.639
No uncertainty penalty	19.44 $\pm$ 0.24	0.28	0.690
Global top- k	18.64 $\pm$ 0.38	1.08	0.684
Uniform clusters	19.21 $\pm$ 0.28	0.51	0.709

also contribute.

Cross-PEFT transfer. Table 12 supports the introduction’s 2.0-point cross-family mismatch penalty. Performance falls as the source PEFT moves farther from the target PEFT. Figure 4 shows the same mechanism from the selection side—PCU-Select’s subset overlap falls as configurations diverge, while PEFT-agnostic RDS+ remains exactly 1.000.

Unseen-configuration transfer. Table 13 is the paired source for Figure 3. Near-support targets transfer zero-shot. Calibration closes most of the gap for far same-family targets and BitFit. Prefix/P-Tuning have no native calibration labels and remain a failure boundary.

Table 12: Cross-PEFT transfer matrix (motivation study). Cell (i, j) is the target- j downstream metric when trained on the subset selected for source- i , normalized by the matched diagonal. Off-diagonal cells fall below 1.00, and the two directions of a mismatched pair differ by 0.01–0.05 rather than a constant offset. Averaged over all mismatched source-target pairs the gap is 1.87 absolute points (mean matched 21.4 vs. mismatched 19.53 on the GSM8K-scale probe). Cross-family mismatches average a 2.0-point gap, the penalty cited in the introduction. Columns are target PEFTs, rows are the source PEFT used for selection.

Source ↓ / Target →	L-r8-qv	L-r16-qkvo	IA3-attnmlp	AD-b64
L-r8-qv	1.00	0.96	0.87	0.93
L-r16-qkvo	0.94	1.00	0.91	0.88
IA3-attnmlp	0.89	0.86	1.00	0.95
AD-b64	0.90	0.92	0.94	1.00

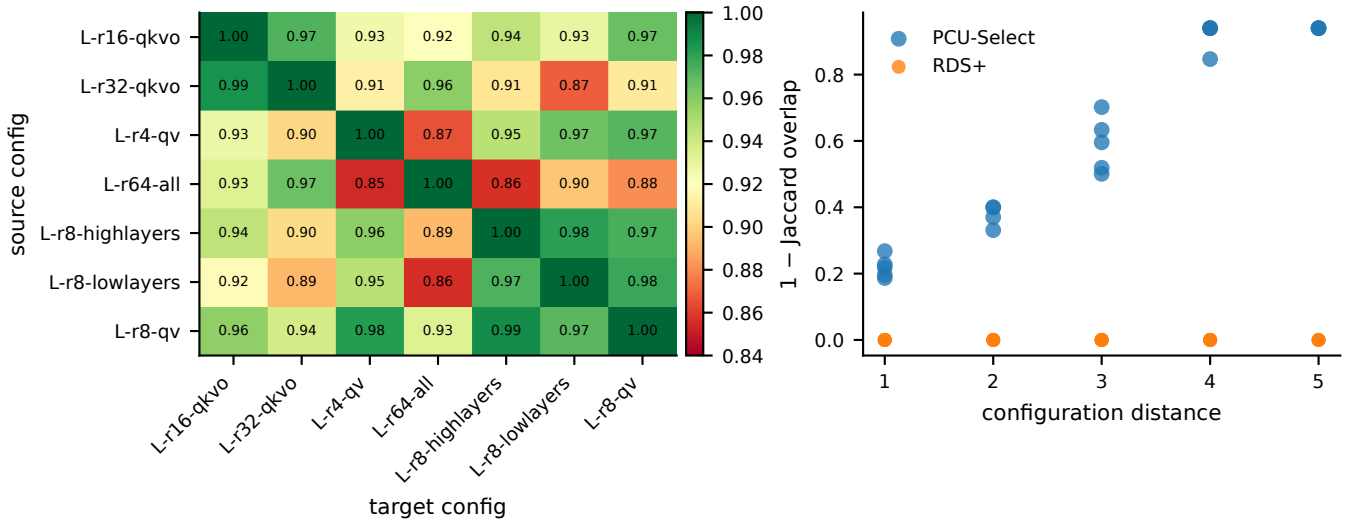


Figure 4: PCU-Select reacts to configuration changes. **Left:** target performance after training on a source-selected subset, normalized by the matched diagonal. The mean off-diagonal drop of 0.068 corresponds to a 1.45-point gap (21.25 matched vs. 19.79 mismatched). **Right:** selection difference ($1 - \text{Jaccard}$) versus a heuristic configuration distance combining log-rank, placement, and module-set changes. Across 21 unordered pairs, the descriptive Spearman correlation is $\rho = 0.969$; leave-one-configuration-out values span $[0.947, 0.972]$. RDS+ lies at zero because its same-task ranking is PEFT-independent.

Table 13: Transfer to unseen configurations over GSM8K, HumanEval, and MMLU. Each entry is PCU-Select minus the named target-specific reference, reported as paired mean $\pm$ sample SD over three target-training seeds. Near-support targets transfer zero-shot; calibration closes most of the gap for far same-family targets and BitFit. Prefix/P-Tuning lack compatible short-update labels and remain a zero-shot failure boundary.

Tier	Targets	Ref.	Zero-shot	Cal-200	Cal-500
L0	L-r32-qkvo, AD-b16	LESS	-0.34 $\pm$ 0.41	—	—
L1	L-r64-all, AD-b256, L-r8-highlayers	LESS	-2.08 $\pm$ 0.44	-0.91 $\pm$ 0.29	-0.30 $\pm$ 0.36
L2-BF	BF	LESS	-4.08 $\pm$ 0.70	-0.65 $\pm$ 0.26	-0.23 $\pm$ 0.42
L2-Prompt	PRE-l16, PT-l32	RDS+	-6.97 $\pm$ 0.34	—	—